EXAM SOLUTIONS

1. a) i) True. $(i+1)$th order stationary guarantees ith order stationarity, but not vice versa. [2]

ii) False. Conditioning on an event Y may increase or decrease the probability of event Z, and it remains the same if they are independent. [2]

iii) False. Conditioning on an event Y decreases the entropy of a random variable X, while it remains the same if they are independent. [2]

iv) False. All white Gaussian processes are wide-sense stationary, but the opposite is not true. A counter example is a low-pass filtered white Gaussian process. This process remains Gaussian (linear filtering preserves Gaussianity), and wide-sense stationary (its statistical properties do not change with time), but it is no longer white (its power spectral density is no longer flat, and samples become correlated). [2]

v) False. The Wiener-Khintchin relation holds only for wide-sense stationary processes, so it may not hold for a first-order stationary process if its autocorrelation function is time-varying. [2]

vi) True. Since integration is a linear transform, the output is WSS and Gaussian, and hence, SSS. [2]

b) i) From the Nyquist theorem, the minimum sampling rate is $2f_c$. [1]

ii) Given that the signal is $m(t) = C \cdot \cos(2\pi f_c t + \Theta)$, its power is found as $P_S = \frac{1}{2}C^2$. [2]

iii) Note that the DC offset D does not affect the quantization noise power because it just shifts the signal but does not change its dynamic range.

We should have $\Delta = \frac{2.4C}{2^{n+1}}$.

Then, the noise power is given by $P_N = \frac{\Delta^2}{12} = \frac{0.48 \cdot C^2}{4^{n+1}}$. [2]

iv) We have

$$SNR = \frac{P_S}{P_N} = \frac{4^{n+1}}{0.96} = 4.16 \cdot 2^{2n}.$$

[2]

v) If $10\log_{10}(SNR) = 67.78$ dB, then we have $SNR = 10^{67.78/10} \approx 10^{6.778} \approx 6,000,000$.

Then, we need $4.16 \times 2^{2n} \geq 6,000,000$. Equivalently, $2^{2n} \geq 1442308$, and $2n \geq \log_2(1442308) \approx 20.46$. This results in $n \geq 10.23$. Hence, the smallest $n = 11$. [2]

c) Since $\boldsymbol{Y}$ is an affine function of $\boldsymbol{X}$, it is also a Gaussian random variable. There-

fore, we only need to identify its mean vector and covariance matrix.

We have

$$\begin{aligned} E[\boldsymbol{Y}] =& E[\boldsymbol{A}\boldsymbol{X} - \boldsymbol{b}], \\ =& \boldsymbol{A}E[\boldsymbol{X}] - \boldsymbol{b}, \\ =& \boldsymbol{A}\boldsymbol{\mu} - \boldsymbol{b}. \end{aligned}$$

The covariance matrix of $\boldsymbol{Y}$ is given by

$$\begin{aligned} E[(\boldsymbol{Y} - E[\boldsymbol{Y}])(\boldsymbol{Y} - E[\boldsymbol{Y}])^T] &= E[\boldsymbol{Y}\boldsymbol{Y}^T] - E[\boldsymbol{Y}]E[\boldsymbol{Y}]^T \\ &= A\boldsymbol{\Sigma}A^T \ . \end{aligned}$$

[3]

d) i) The capacity of a BSC is given by $C = 1 - H(p)$, which for $p = 0.3$ and $p = 0.8$, gives us $C = 0.118$ and $C = 0.278$ bits per channel use, respectively. [3]

ii) The capacity is higher for $p = 0.8$. A channel with a very high error probability (like $p = 0.8$) can be more useful than one with a moderate error probability (like $p = 0.3$) because the high predictability of errors allows for better error correction. [3]

e) i) Noise power is given by

$$\int_{-\infty}^{\infty} S_n(f)df = 2 * 200 + 1 * 200 = 600.$$

[2]

ii) This corresponds to baseband communication. The signal power is given by $P_m = \int_{-\infty}^{\infty} S_m(f)df = 1$.

The noise power depends on W.

If $W = 200$, we have $P_n = 600$, and $SNR = 1/600$. [2]

iii) This corresponds to passband communication with DSB-SC. The signal power is still the same, $P_m = 1$.

The noise power depends on W, and is given by $2N_0W$, where $N_0/2 = 1$ if $f_2 = 150$ and where $N_0/2 = 2$ if $f_2 = 50$.

If $f_1 = f_2 = 50$ and $W = 5$, we have $P_n = 40$, and $SNR = 1/40$. [3]

iv) This corresponds to a phase shift of $\pi/2$ at the receiver. This means that the receiver demodulates the in-phase component instead of the quadrature component, on which the information is modulated. Therefore, there will be no information at the receiver, and hence, $SNR = 0$. [3]

2. a) i) The matched filter has impulse response $h(t) = q(T-t)$. Hence the output at time T is

$$y(T) = \int_0^T r(\tau)d\tau = A_i T + \int_0^T n(\tau)d\tau$$

Since $\int_0^T n(\tau)d\tau$ is a Gaussian with zero-mean and variance $\frac{N_0 T}{2}$, we have

$$y(T) \sim N(A_i T, \frac{N_0 T}{2}).$$

[3]

ii) The probability of error for each message is given by

$$P_{e,1} = \int_{-\infty}^{-\lambda} \frac{1}{\sqrt{\pi N_0 T}} e^{-(x+AT)^2/N_0 T} dx = Q\left(\frac{AT-\lambda}{\sqrt{N_0 T/2}}\right)$$
$$P_{e,3} = \int_{\lambda}^{+\infty} \frac{1}{\sqrt{\pi N_0 T}} e^{-(x-AT)^2/N_0 T} dx = Q\left(\frac{AT-\lambda}{\sqrt{N_0 T/2}}\right)$$

and

$$P_{e,2} = 2\int_{\lambda}^{+\infty} \frac{1}{\sqrt{\pi N_0 T}} e^{-x^2/N_0 T} dx = 2Q\left(\frac{\lambda}{\sqrt{N_0 T/2}}\right)$$

[6]

iii) The total probability of error under threshold λ is given by

$$P_e = \sum_{i=1}^{3} p_i P_{e,i} = (p_1+p_3)Q\left(\frac{1-\lambda}{\sqrt{1/2}}\right) + 2p_2 Q\left(\frac{\lambda}{\sqrt{1/2}}\right)$$

The optimal threshold is found by solving $\frac{dP_e}{d\lambda} = 0$. We use the Leibnitz rule to take the derivative, and obtain the following:

$$\frac{dP_e}{d\lambda} = -(p_1+p_3)\frac{1}{\sqrt{\pi}} e^{-(1-\lambda)^2} + 2p_2 \frac{1}{\sqrt{\pi}} e^{-\lambda^2} = 0.$$

It follows that

$$\ln \frac{p_1+p_3}{2p_2} = (1-\lambda)^2 - \lambda^2 = 1-2\lambda.$$

Given that $p_2 = 1/2$, we immediately get that the optimal threshold is $\lambda^* = \frac{1}{2}(1-\ln\frac{1}{2})$. Under this detection threshold, the total probability of error is

$$P_e = \frac{1}{2}Q\left(\frac{1-\lambda^*}{\sqrt{1/2}}\right) + Q\left(\frac{\lambda^*}{\sqrt{1/2}}\right).$$

[4]

iv) We should choose signals $s_1(t)$ and $s_3(t)$. For example, transmit $s_1(t)$ when $j=1$ and $s_3(t)$ when $j=2$. The optimal detection rule still has a threshold structure. Assume the threshold is λ, so that the decision for "1" is made if the matched filter output is below λ; otherwise decision for "2" is made. The probabilities of error are given by

$$P_{e,1} = \int_{-\infty}^{\lambda} \frac{1}{\sqrt{\pi N_0 T}} e^{-(x+AT)^2/N_0 T} dx = Q\left(\frac{AT+\lambda}{\sqrt{N_0 T/2}}\right)$$
$$P_{e,2} = \int_{\lambda}^{+\infty} \frac{1}{\sqrt{\pi N_0 T}} e^{-(x-AT)^2/N_0 T} dx = Q\left(\frac{AT-\lambda}{\sqrt{N_0 T/2}}\right).$$

Using a similar argument as before, it is easy to see that the optimal threshold is $\lambda^* = 0$.

[3]

b) i) The coordinates are $(A,A),(A,-A),(-A,A)$ and $(-A,-A)$. The energy per symbol is $E_s = I^2 + Q^2 = 2A^2$. [2]

ii) For 4-QAM, a symbol error occurs if either the I or Q component (or both) is incorrect. In each dimension, the noise is Gaussian with mean 0 and variance $N_0/2$. Hence the probability of error in one dimension is

$$P_{e,I} = P_{e,Q} = Q\left(\frac{A}{\sqrt{N_0/2}}\right).$$

The symbol error probability can be found from the probability of correct detection, which is given by the probability that no error occurs in either I or Q component; hence, it is given by

$$P_s = 1-(1-P_{e,I})(1-P_{e,Q}) = 1-\left[1-Q\left(\sqrt{2A^2/N_0}\right)\right]^2.$$

[4]

iii) With a phase error ϕ, the receiver's local oscillator is $\cos(2\pi f_c t+\phi)$ and $\sin(2\pi f_c t+\phi)$. After demodulation and low-pass filtering, the transmitted symbol (A,A) at the receiver becomes

- I-component: $A\cos(\phi)+n$
- Q-component: $A\cos(\phi)+n$

where n denotes the noise. The desired signal is rotated, reducing the effective amplitude in the correct dimension.

[4]

iv) The effective distance to the decision boundary is reduced to $A\cos(\phi)$. Hence the symbol error probability becomes

$$P_s = 1-(1-P_{e,I})(1-P_{e,Q}) = 1-\left[1-Q\left(\sqrt{2A^2\cos^2(\phi)/N_0}\right)\right]^2.$$

Clearly, the phase error increases the symbol error probability.

3. a) i) We can do this by following the Huffman procedure. One such resultant code is given by codewords $a \to 101, b \to 11, c \to 00, d \to 100$, and $e \to 01$.

This code is not unique. We can break ties in the Huffman procedure arbitrarily and obtain codes with the same average length.

The average codeword length is 2.25 bits per symbol. [5]

ii) Entropy of this source is 2.198 bits per source symbol. It can be calculated as

$$\begin{aligned} & 0.1 \cdot \log(10) + 2 \cdot 0.15 \cdot \log(20/3) + 2 \cdot 0.3 \cdot \log(10/3) \\ =& 0.1 \cdot (1 + 2.32) + 0.3 \cdot (2 + 2.32 - 1.58) + 0.6 \cdot (1 + 2.32 - 1.58) \\ =& 2.198 \end{aligned}$$

[3]

iii) The compression efficiency can be improved by creating larger blocks of symbols and applying the Huffman procedure to these blocks. The theoretical limit for compression efficiency is the entropy of the source, so we can at most achieve a reduction of

$(2.25 - 2.198)/2.25 = 0.023 = 2.3\%$

in compression rate. [3]

iv) The average codeword length is 2.45 bits per source symbol. [3]

v) This can be considered as a new source U whose probabilities are the averages of the probabilities of S and T: $\Pr(U = a) = 0.2$, $\Pr(U = b) = 0.2$, $\Pr(U = c) = 0.25$, $\Pr(U = d) = 0.15$, and $\Pr(T = e) = 0.2$.

If we use the original code, the average codeword length will be 2.35 bits per symbol.

If we apply the Huffman procedure on the new mixture source U, we get the following code $a \to 01, b \to 10, c \to 11, d \to 000$ and $e \to 001$, which has an average codeword length of 2.35 bits per symbol. In this particular case, there is no gain from designing a new code. [3]

b) i) We can have 1 outcome of length 2, AA, with probability $(1/2)^2$.

We can have 3 outcomes of length 3, ALA, LAA, LLL, each with probability $(1/2)^3$.

We can have 6 outcomes of length 4, ALLL, LALL, LLAL, ALLA, LLAA, LALA, each with probability $(1/2)^4$.

There can be no outcome with 5 or more plays.

Then we have

$$\begin{aligned} H(X) &= \frac{1}{4}\log 4 + 3\frac{1}{8}\log 8 + 6\frac{1}{16}\log 16 \\ &= 3.125 \end{aligned}$$

[4]

ii) The probability of observing a sequence of length 2 is $1/4$.

The total probability of observing a sequence of length 3 is $3/8$.

The total probability of observing a sequence of length 4 is $6/16$.

$$\begin{aligned} H(Y) &= \frac{1}{4}\log 4 + 2 \cdot \frac{3}{8}\log(8/3) \\ &= 1.562 (\sim 1.565) \end{aligned}$$

[3]

iii) Since Y is a deterministic function of X, we have $H(Y|X) = 0$. [2]

iv) Since $H(X,Y) = H(X) + H(Y|X) = H(Y) + H(X|Y)$, we get $H(X|Y) = H(X) + H(Y|X) - H(Y) = 3.125 - 1.562 = 1.563$. [4]